

GPU Saturation Scorer Report

Generated on 2026-03-31 13:28:09

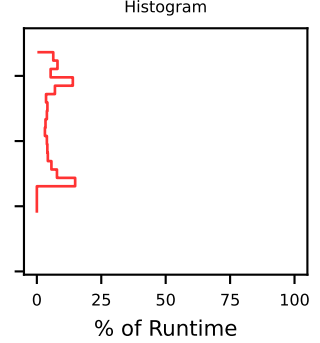
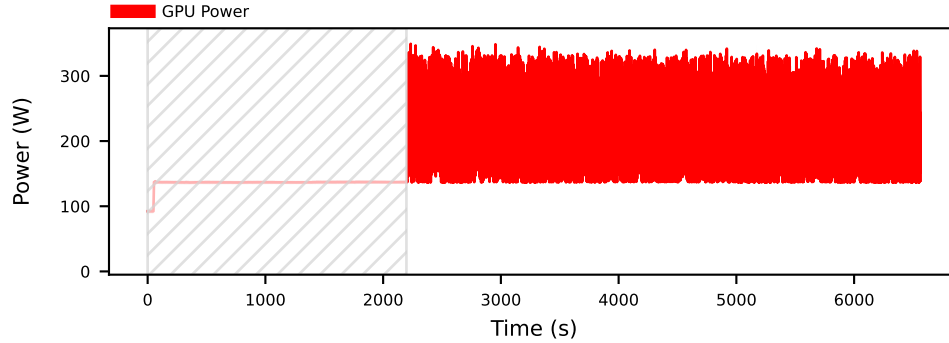
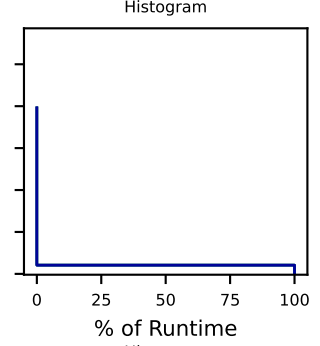
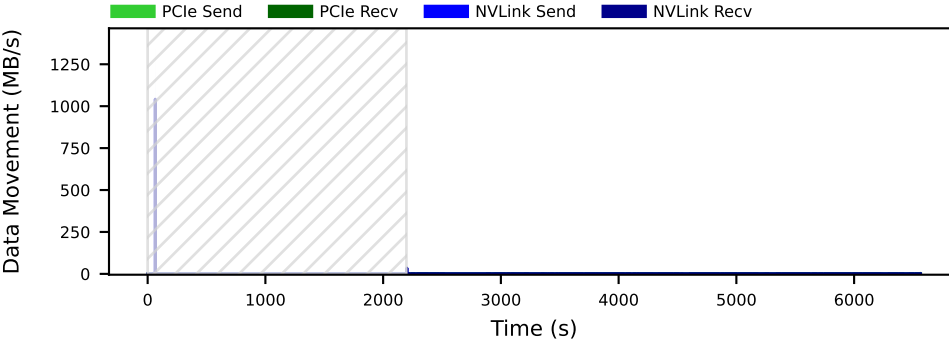
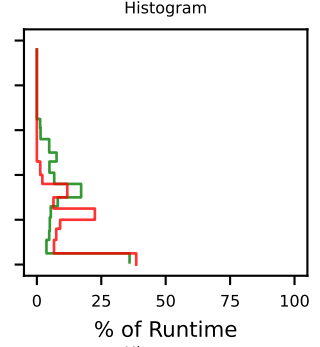
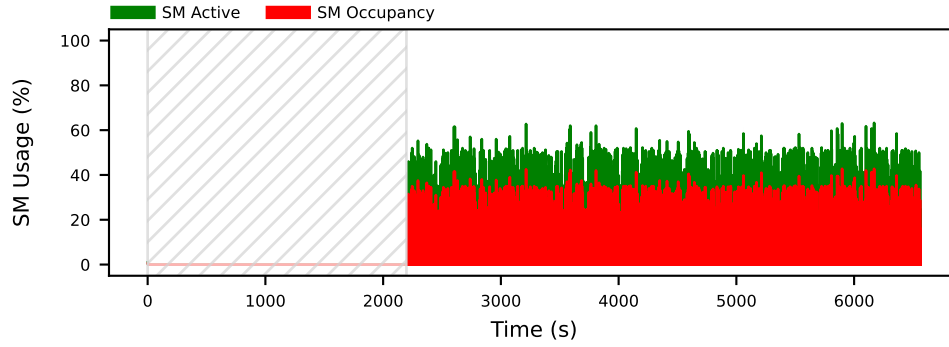
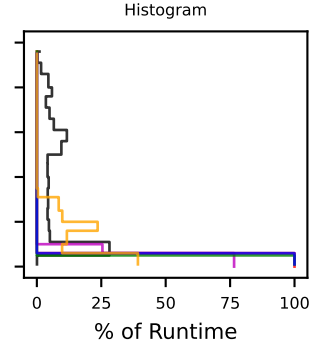
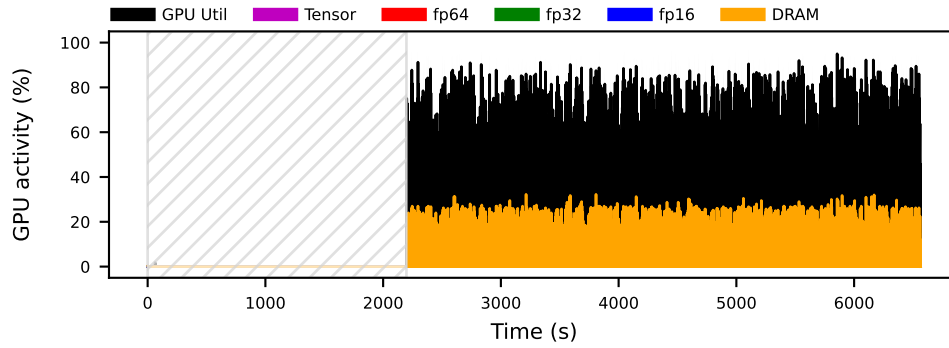
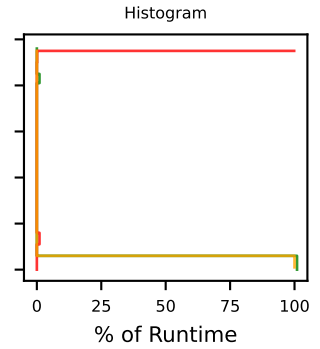
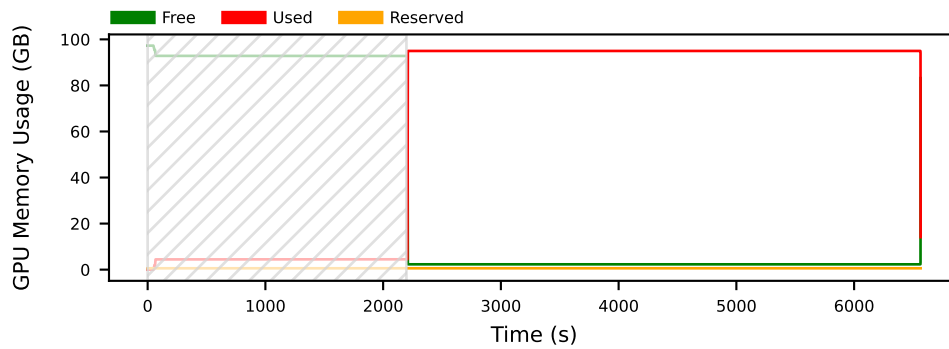
Slurm Job ID	1766781
Job Name	OneAstro-bench
Jobstep	0
Cluster	clariden
Date	2026-03-31T08:03:03+0200
Node Count	1 (of 1)
Task Count	1 (of 1)
GPU Count	4
GPU Energy Use	1165.31 Wh
Executable	torchrun
Arguments	'--nnodes=1' '--node_rank=0' '--nproc_per_node=4' '--rdzv_backend=c10d' '--rdzv_endpoint=nid006686:29500' './train_aion_ddp_v4.12_token.py' '--model_name' 'polymathic-ai/aion-base' '--batch_size' '256' '--weight_decay' '0.1' '--num_workers' '16' '--use_amp' '--use_tensorboard' '--grad_accum_steps' '1' '--num_workers' '16' '--input_budget' '256' '--anchor_ratio_min' '0.8' '--anchor_ratio_max' '1.6' '--output_budget' '128' '--beta_alpha' '1.0' '--beta_beta' '4' '--data_path' '/capstor/store/cscs/pasc/c39/swiss-ai/test/data/train/' '--output_dir' '/capstor/store/cscs/pasc/c39/swiss- [...]

Evaluation (Experimental)

Criteria	Total Runtime	After GPU Util > 50%
GPU Start ≤ 0 - poor ≤ 25 - improve ≤ 50 - acceptable ≤ 100 - good	66.53 % (good) The GPU usage began very early on during the measurement.	
GPU Util ≤ 25 - poor ≤ 50 - improve ≤ 75 - acceptable ≤ 100 - good	25.89 % (improve) The global average GPU utilization is below acceptable limits.	28.28 % (improve) The global average GPU utilization is below acceptable limits.
FP Util ≤ 0.0625 - poor ≤ 0.125 - improve ≤ 0.25 - acceptable ≤ 4 - good	0.02 (poor) The global average FP utilization is poor and below acceptable use of node resources.	0.03 (poor) The global average FP utilization is poor and below acceptable use of node resources.
Power ≤ 100 - poor ≤ 200 - improve ≤ 500 - acceptable ≤ 700 - good	200.43 W (acceptable) The global average power draw is reasonable and indicates good GPU activation.	206.66 W (acceptable) The global average power draw is reasonable and indicates good GPU activation.

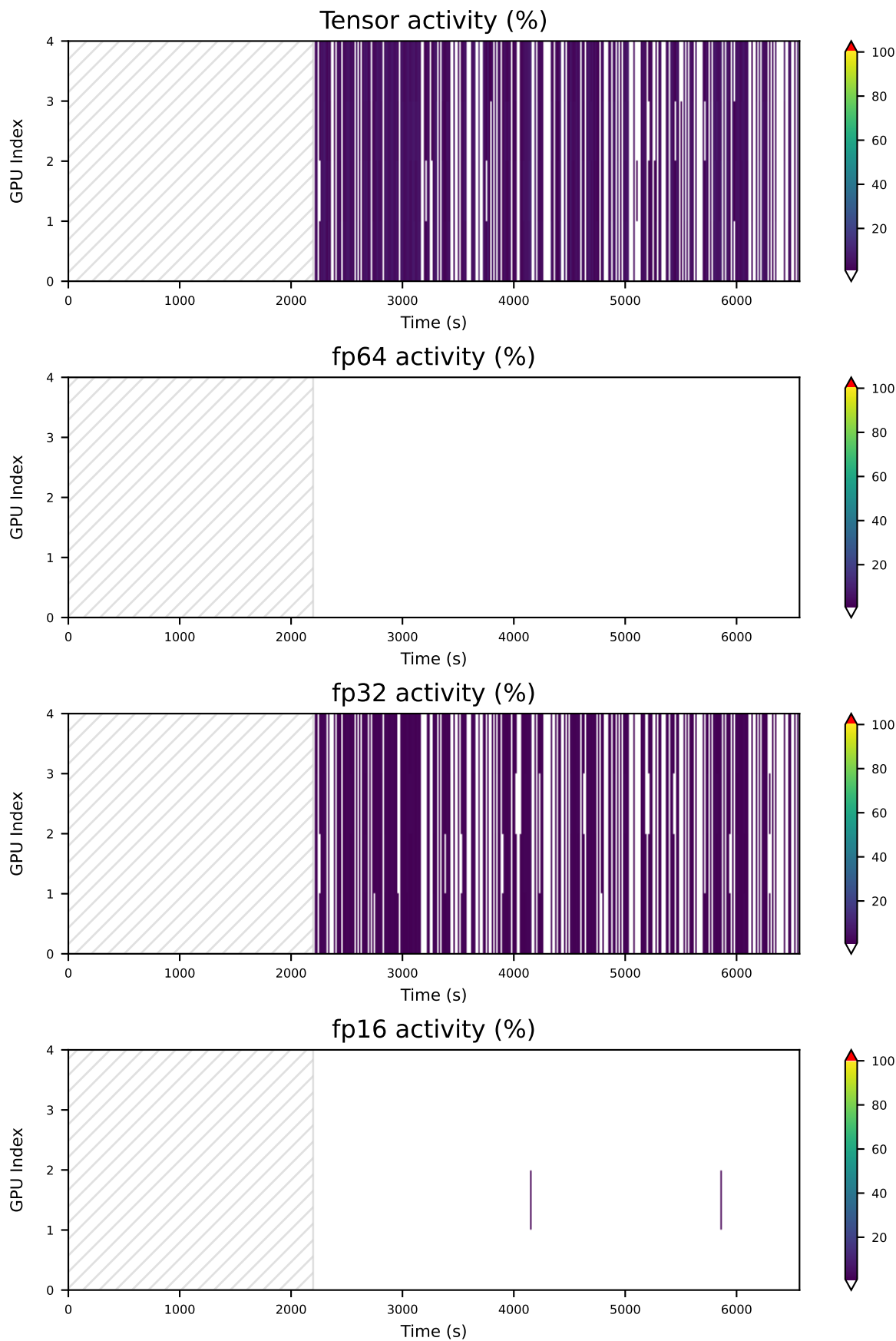
Aggregate GPU Performance Metrics

Data summarized over all GPUs.



Individual GPU Performance Metrics

Data from all GPUs.



Definitions

From Nvidia Documentation

GPU Memory (MB) Used/Free/Reserved	Used/Free/Reserved Frame Buffer of the GPU in MB
GPU Utilization (%)	The fraction of time any portion of the graphics or compute engines were active. The graphics engine is active if a graphics/compute context is bound and the graphics/compute pipe is busy. The value represents an average over a time interval and is not an instantaneous value.
SM Active (%)	The fraction of time at least one warp was active on a multiprocessor, averaged over all multiprocessors. Note that "active" does not necessarily mean a warp is actively computing. Warps waiting on memory requests are considered active. The value represents an average over a time interval and is not an instantaneous value.
SM Occupancy (%)	The fraction of resident warps on a multiprocessor, relative to the maximum number of concurrent warps supported on a multiprocessor. The value represents an average over a time interval and is not an instantaneous value. Higher occupancy does not necessarily indicate better GPU usage. For GPU memory bandwidth limited workloads, higher occupancy is indicative of more effective GPU usage. However if the workload is compute limited, then higher occupancy does not necessarily correlate with more effective GPU usage.
Tensor Active (%)	The fraction of cycles the tensor (HMMA / IMMA) pipe was active. The value represents an average over a time interval and is not an instantaneous value. Higher values indicate higher utilization of the Tensor Cores. An activity of 100% is equivalent to issuing a tensor instruction every other cycle for the entire time interval.
FP Active (%) for FP64, FP32, FP16	The fraction of cycles the respective float-point pipe was active. The value represents an average over a time interval and is not an instantaneous value. Higher values indicate higher utilization of the floating-point cores.
DRAM Active (%)	The fraction of cycles where data was sent to or received from device memory. The value represents an average over a time interval and is not an instantaneous value. Higher values indicate higher utilization of device memory. An activity of 100% is equivalent to a DRAM instruction every cycle over the entire time interval (in practice a peak of ~0.8 (80%) is the maximum achievable).
NVLink Send/Recv Bandwidth (byte/s)	The rate of data transmitted / received over NVLink, not including protocol headers. The theoretical maximum NVLink Gen2 bandwidth is 25 GB/s per link per direction.
PCIe Send/Recv Bandwidth (byte/s)	The rate of data transmitted / received over the PCIe bus, including both protocol headers and data payloads. The theoretical maximum PCIe Gen3 bandwidth is 985 MB/s per lane.